EXPERIMENT :- 2

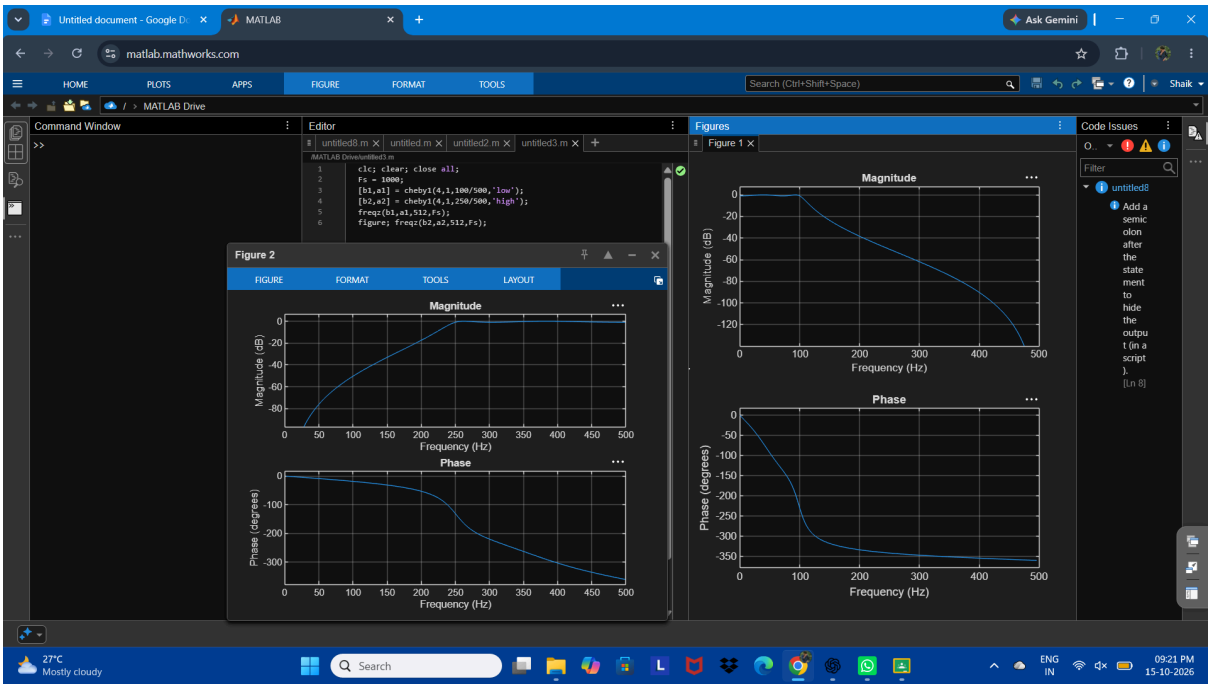
PROGRAM CODES ➖

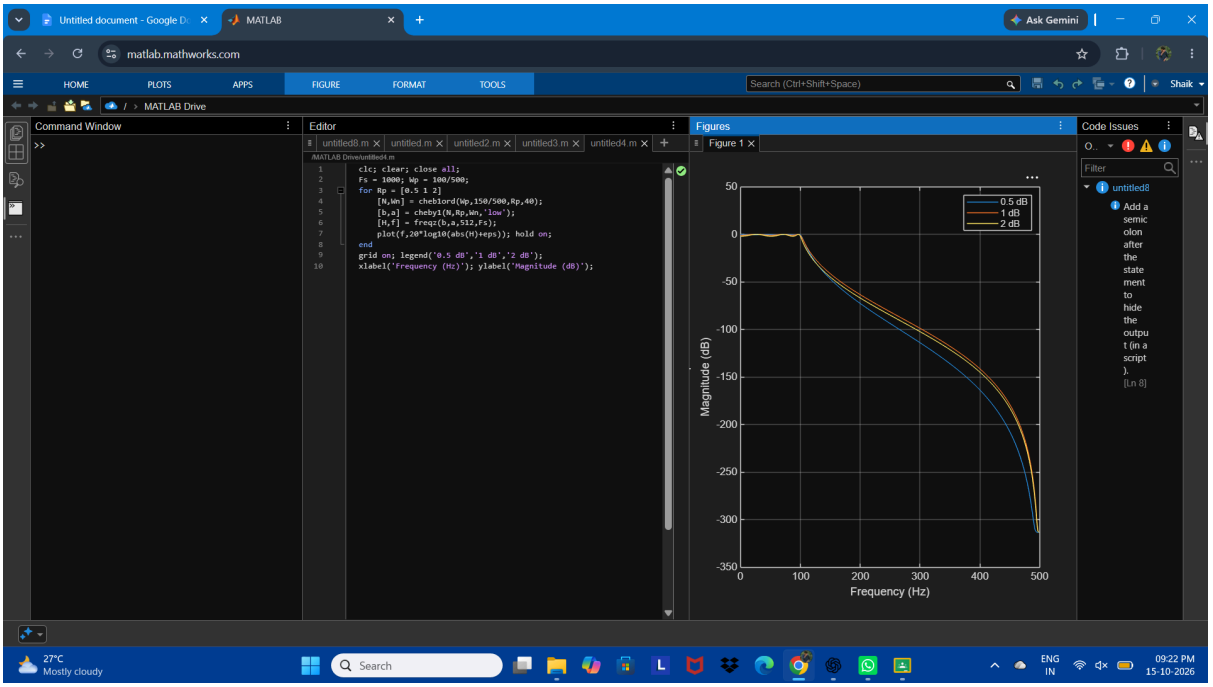
```
clc; clear; close all;
Fs = 1000;
[b1,a1] = cheby1(4,1,100/500,'low');
[b2,a2] = cheby1(4,1,250/500,'high');
freqz(b1,a1,512,Fs);
figure; freqz(b2,a2,512,Fs);
```

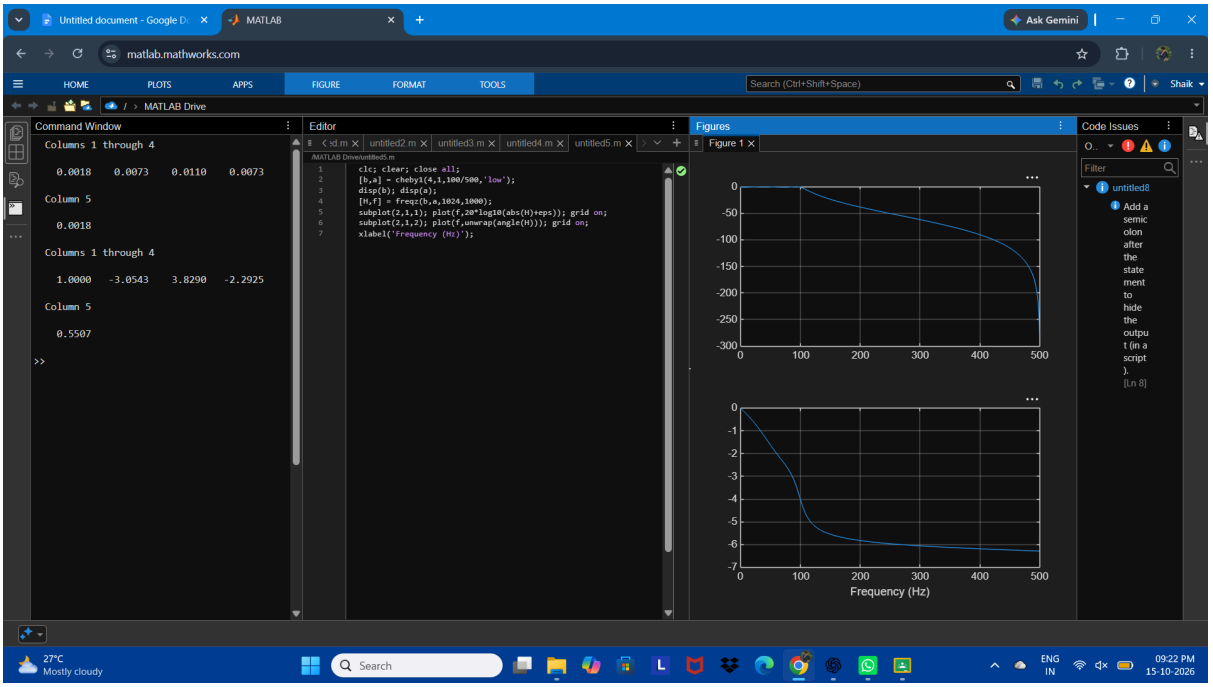
```
clc; clear; close all;
Fs = 1000; Wp = 100/500;
for Rp = [0.5 1 2]
   [N,Wn] = cheb1ord(Wp,150/500,Rp,40);
   [b,a] = cheby1(N,Rp,Wn,'low');
   [H,f] = freqz(b,a,512,Fs);
   plot(f,20*log10(abs(H)+eps)); hold on;
end
grid on; legend('0.5 dB','1 dB','2 dB');
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
```

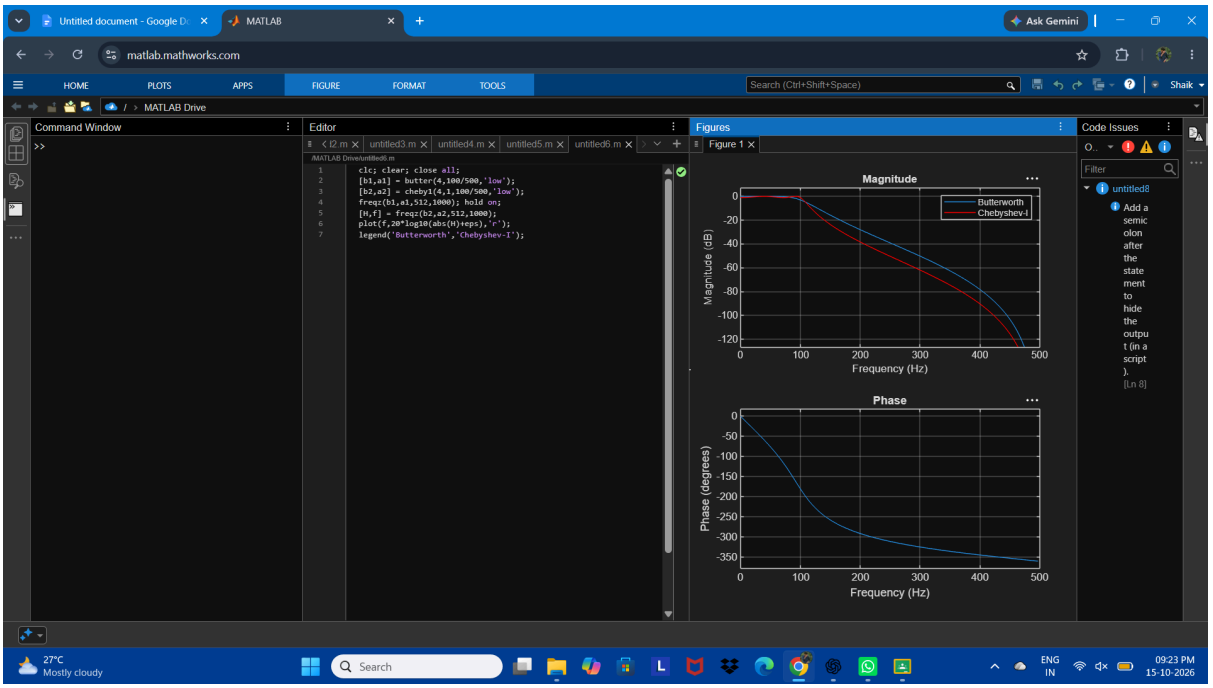
```
clc; clear; close all;
[b,a] = cheby1(4,1,100/500,'low');
disp(b); disp(a);
[H,f] = freqz(b,a,1024,1000);
subplot(2,1,1); plot(f,20*log10(abs(H)+eps)); grid on;
subplot(2,1,2); plot(f,unwrap(angle(H))); grid on;
xlabel('Frequency (Hz)');
```

```
clc; clear; close all;
[b1,a1] = butter(4,100/500,'low');
[b2,a2] = cheby1(4,1,100/500,'low');
freqz(b1,a1,512,1000); hold on;
[H,f] = freqz(b2,a2,512,1000);
plot(f,20*log10(abs(H)+eps),'r');
legend('Butterworth','Chebyshev-I');
```






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clc; clear; close all;
[b,a] = cheby1(4,1,100/500,'low');
disp(b); disp(a);
[H,f] = freqz(b,a,1024,1000);
subplot(2,1,1); plot(f,20*log10(abs(H)+eps)); grid on;
subplot(2,1,2); plot(f,unwrap(angle(H))); grid on;
xlabel('Frequency (Hz)');
Figures
Figure 1
Frequency (Hz)
Code Issues
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Add a semicolon after the statement to hide the output (in a script).
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clc; clear; close all;
[b1,a1] = butter(4,100/500,'low');
[b2,a2] = cheby1(4,1,100/500,'low');
freqz(b1,a1,512,1000); hold on;
[H,f] = freqz(b2,a2,512,1000);
plot(f,20*log10(abs(H)+eps),'r');
legend('Butterworth','Chebyshev-I');
Figures
Figure 1
Magnitude
Butterworth
Chebyshev-I
Magnitude (dB)
Frequency (Hz)
Phase
Phase (degrees)
Frequency (Hz)
Code Issues
Filter
untitled8
Add a semicolon after the statement to hide the output (in a script).
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